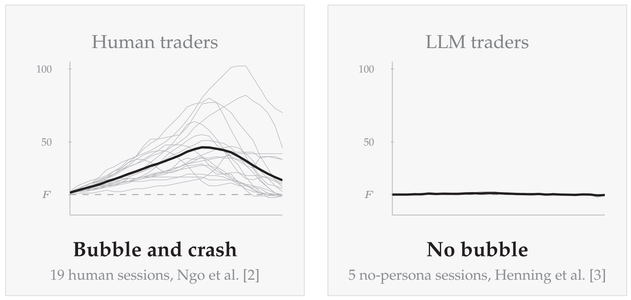
Building Traders: Persona Construction for LLM Agents in Experimental Asset Markets

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The Problem

- Laboratory asset markets bubble and crash reliably, though at hundreds to thousands of dollars per session [1, 2].
- LLM markets cost far less, and we can choose which traders are in them. But an agent has no disposition until given one.



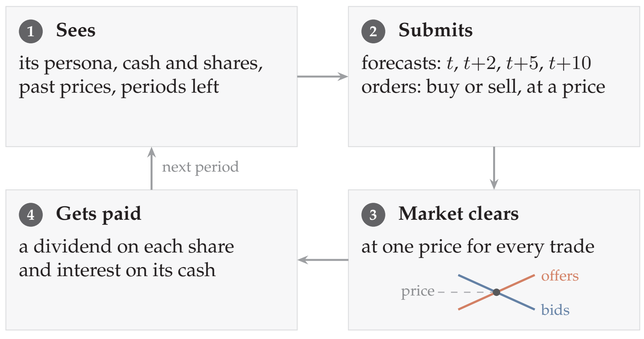
NOW **Can personas make agent markets behave like human ones?**

NEXT **Which behavioral traits create bubbles?**

The Market

- Call market, T=30 periods, 12 `claude-sonnet-5` agents.
- Each agent starts with \$100 and 4 shares; cash earns 5% a period.
- A share pays \$0.40 or \$1.00, 50/50, and is bought back for \$14 at the end, so $F = \mathrm{E}[D]/r = \$14$ in every period.

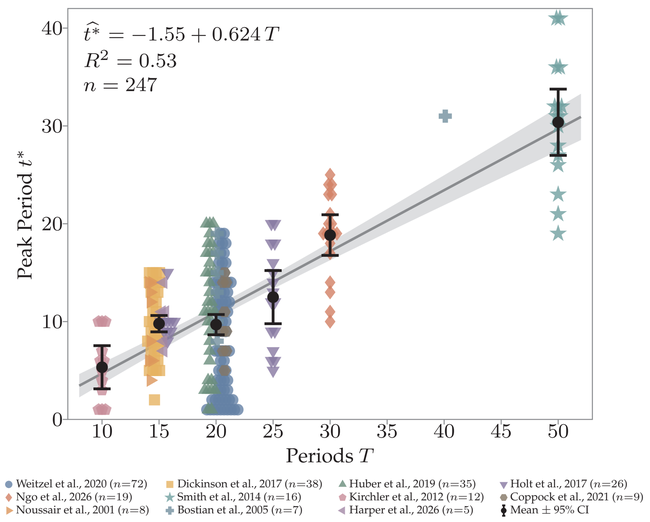
Design after Smith et al. [4]. Flat fundamental value of Bostian et al. [5].



One period from the agent's side. It never sees the order book.

The Benchmark: Peak Timing

A bubble's *size* is study-specific, from 1.01 to 10.04 times fundamental value. Its *timing* is not.



Every lab session we could find with a constant fundamental value, one mark per session, jittered within a study. Gray band: 95% CI on the fit.

Studies that agree on little still agree on when the peak arrives.

Theory-Derived Archetypes

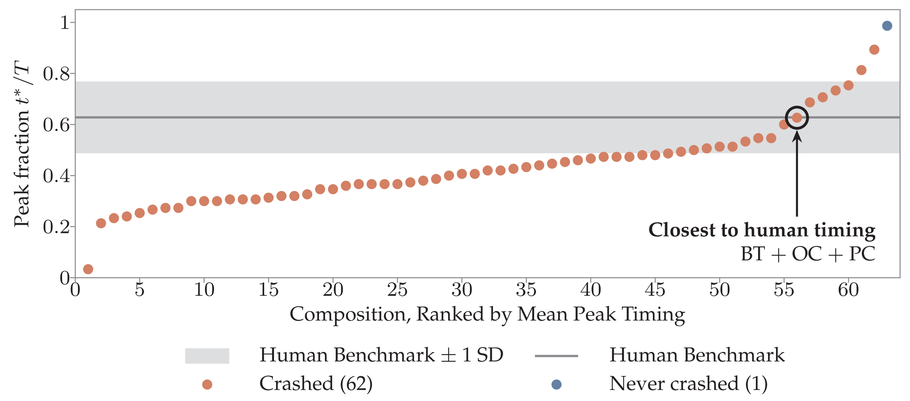
Written from theory, with no human data.

	Archetype	Decision summary
RA	Rational Arbitrageur	buys on expected return above a threshold
MC	Momentum Chaser	buys rising prices, sells on the dip
BT	Bubble Timer	rides up, exits when the rise decelerates
PC	Precommitter	fixes buy and sell periods in advance
NT	Noise Trader	vague prior, updates inconsistently
OC	Overconfident Contrarian	buys falling prices, never sells losers

Full prompts, a paragraph apiece, in appendix.

Every non-empty subset is a population: 63 compositions, 315 sessions.

The composition is the design variable, not any one prompt.

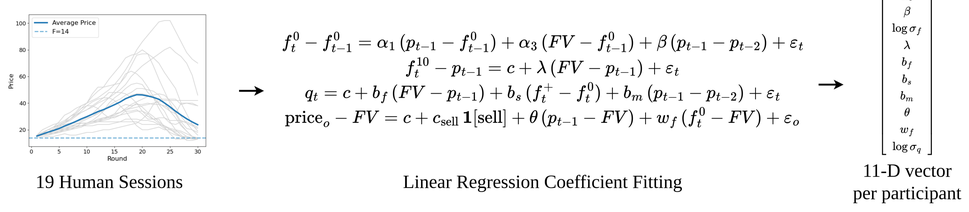


One point per composition, ranked. Band: human mean ± 1 SD. Composition means run from 0.03 to 0.99.

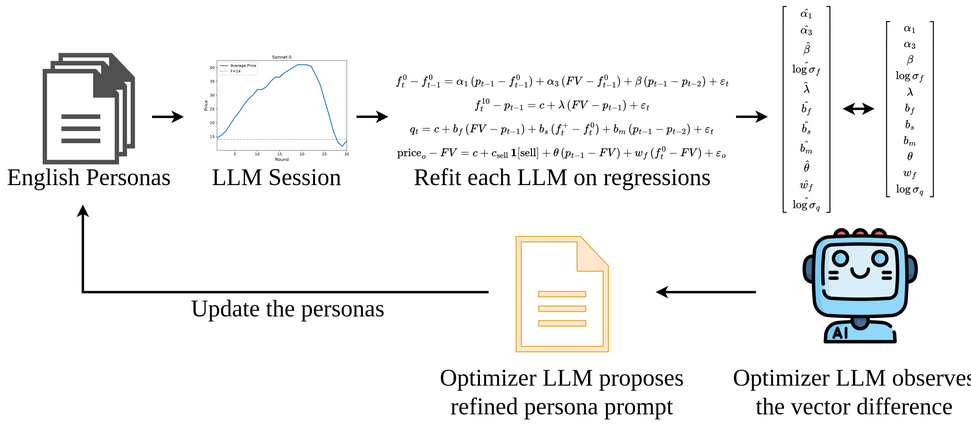
Change the mix of traders and you change when the bubble peaks.

Evolved Personas

Fit to coefficients estimated from 224 human subjects [2]. The three types are not equally common, and the markets here keep the human proportions.



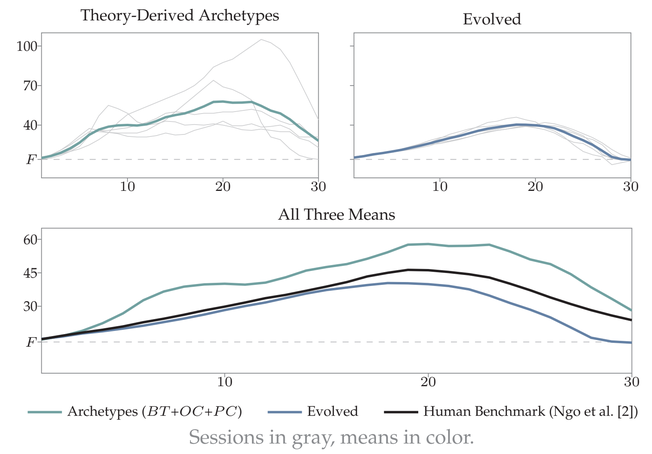
Stage 1. 11 behavioral coefficients per subject by OLS, then a mixture over them recovers three trader types.



Stage 2. After Tao et al. [6]: an optimizer rewrites the persona text until the agents' coefficients match the type means.

Human behavior goes in as numbers and comes back out in English.

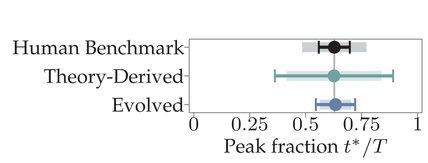
Both Families Bubble



Sessions in gray, means in color.

Agent markets take approximately the same shape as human ones: a bubble, then a crash.

Peak Timing by Population



	Mean	SD	Sessions
Human Benchmark	0.628	0.14	19
Theory-Derived	0.627	0.21	5
Evolved	0.633	0.07	5

Dot: mean. Bar: 95% CI. Band: ±1 SD. Agents at $T = 30$.

Both families peak when human markets do.

Why Agents at All

	Humans	LLM agents
No IRB	×	✓
Low cost	×	✓
Tireless and attentive	×	✓
Full retention in long studies	×	✓

- **Personas make LLM markets bubble like human ones.** Fit only to how humans trade, they peak when human sessions do.
- **But the evolved personas are more alike than people are.** Their peak fractions are only 0.27 as spread out as the human ones, so the average human session is reproduced and the variety of them is not.

Not a substitute for human subjects (yet), but a way to search the design space first.

References

[1] Smith, V. L., Suchanek, G. L. & Williams, A. W. (1988). Bubbles, crashes, and endogenous expectations in experimental spot asset markets. *Econometrica* 56(5), 1119–1151.

[2] Ngo, C. H., Henning, T., Spoon, R., Toma, M., Han, J., Smith, A. & Camerer, C. F. (2026). Skin in the game: electrodermal arousal, expectations, and bubbles in experimental asset markets. *SSRN 7525744*.

[3] Henning, T., Ojha, S. M., Spoon, R. & Han, J. (2025). LLM agents do not replicate human market traders. *arXiv:2502.15800*.

[4] Smith, A., Lohrenz, T., King, J., Montague, P. R. & Camerer, C. F. (2014). Irrational exuberance and neural crash warning signals. *PNAS* 111(29), 10503–10508.

[5] Bostian, A. J., Goeree, J. & Holt, C. A. (2005). Price bubbles in asset market experiments with a flat fundamental value. *Experimental Finance Conference*.

[6] Tao, W., Wu, H. & Wong, W.-F. (2026). SePO: self-evolving prompt agent for system prompt optimization. *arXiv:2606.04465*.

Acknowledgments

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Working paper